DHIRUBHAI AMBANI UNIVERSITY

High Performance Computing

Assignment 06

Parallel Scatter-to-Mesh Interpolation

Group 20

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1. Introduction

This report presents our parallel implementation of the scatter-to-mesh interpolation algorithm for HPC Assignment 06. The objective is to parallelise a bilinear (area-weighted) interpolation that maps N randomly distributed scattered points onto a structured Nx × Ny mesh grid, using OpenMP shared-memory parallelism.

The core computational challenge arises from race conditions: multiple scattered points may fall within the same grid cell, causing concurrent write-conflicts on shared mesh grid points. Our implementation resolves this through thread-private mesh privatisation followed by a reduction step.

2. Problem Description

Given a set of N scattered points in a 2D unit domain [0, 1] × [0, 1]:

$$S = \{ (x_i, y_i, f_i) \mid i = 1, 2, \ldots, N \} \quad \text{where } f_i = 1$$

each point must distribute its value to the four surrounding structured mesh grid points via bilinear interpolation weights. The structured mesh G consists of (Nx+1) × (Ny+1) grid points with uniform spacing:

$$\Delta x = 1/Nx, \quad \Delta y = 1/Ny$$

The interpolation accumulates weighted contributions at each grid node. Because fi = 1 for all particles, the grid values represent density-weighted area coverage.

Config	Nx	Ny	Grid Size	Particles	Max Iter
A	250	100	251 × 101	0.9 M	10
B	250	100	251 × 101	5 M	10
C	500	200	501 × 201	3.6 M	10
D	500	200	501 × 201	20 M	10
E	1000	400	1001 × 401	14 M	10

Table 1: Input configurations used for benchmarking

3. Serial Baseline Implementation

The serial implementation iterates over all N scattered points. For each point, it:

1. Computes the enclosing grid cell indices (grid_i, grid_j) using fast floating-point division.
2. Calculates bilinear fractional offsets (frac_x, frac_y) within the cell.
3. Derives four area-proportional weights scaled by the cell area Δx×Δy.
4. Atomically accumulates contributions into the four surrounding mesh nodes.

```
// Serial interpolation kernel
for (int idx = 0; idx < NUM_Points; idx++) {
    int gi = (int)(points[idx].x * NX);  // cell column index
    int gj = (int)(points[idx].y * NY);  // cell row index
    double fx = (points[idx].x * NX) - gi;  // x-fraction in cell
    double fy = (points[idx].y * NY) - gj;  // y-fraction in cell
```

```
    double w00 = (1-fx)*(1-fy)*dx*dy;  // bottom-left weight
    double w10 = fx*(1-fy)*dx*dy;      // bottom-right weight
    double w01 = (1-fx)*fy*dx*dy;      // top-left weight
    double w11 = fx*fy*dx*dy;          // top-right weight

    int base = gj * GRID_X + gi;
    mesh[base]          += w00;
    mesh[base+1]        += w10;
    mesh[base+GRID_X]   += w01;
    mesh[base+GRID_X+1]+= w11;
}
```

Listing 1: Serial interpolation kernel (utils_serial.cpp)

The serial code was validated against the provided test dataset (Test_input.bin vs Test_Mesh.out) prior to parallelisation. All subsequent parallel results were compared against the serial output to confirm correctness.

4. Parallel Implementation

4.1 Parallelisation Strategy: Thread-Private Privatisation

The chosen approach is thread-private mesh privatisation. Each OpenMP thread maintains its own private copy of the full mesh grid. Particles are distributed statically across threads using schedule(static). After processing its assigned particles, each thread merges its local mesh into the global mesh inside a critical section.

This avoids expensive atomic operations on every mesh update (which would serialise writes to hot grid cells) in favour of a single parallel accumulation phase followed by one sequential reduction per thread.

4.2 Race Condition Analysis

The fundamental race condition in naive parallelisation occurs when two threads simultaneously write to the same mesh grid node. Because multiple scattered points may reside in the same or neighbouring grid cells, concurrent updates to shared nodes are unavoidable without synchronisation.

Three strategies were considered:

- Atomic operations: Correct, but expensive when many particles share the same grid cell. Serialises writes to hot nodes, reducing effective parallelism.
- Thread-private privatisation (chosen): Each thread operates on an independent copy. Zero synchronisation during the main computation. Merge is O(grid_size) per thread, sequential.
- Domain decomposition: Partition the grid and assign non-overlapping regions to threads. Boundary cells require careful handling; load balance depends on particle distribution.

4.3 Pseudocode

```
PARALLEL INTERPOLATION PSEUDOCODE
==================================
Input:  points[N], global mesh_value[GRID_X*GRID_Y]
Output: mesh_value[] with accumulated interpolated values

1. memset(mesh_value, 0, GRID_X*GRID_Y)

2. PARALLEL REGION (omp parallel)
```

```
   a. Each thread allocates thread_mesh[GRID_X*GRID_Y] = {0}

   b. OMP FOR SCHEDULE(STATIC)
      for idx = 0 to NUM_Points - 1:
          x = points[idx].x;  y = points[idx].y
          gi = clamp(floor(x * NX), 0, NX-1)
          gj = clamp(floor(y * NY), 0, NY-1)
          fx = x*NX - gi;  fy = y*NY - gj

          w00 = (1-fx)*(1-fy)*dx*dy
          w10 = fx*(1-fy)*dx*dy
          w01 = (1-fx)*fy*dx*dy
          w11 = fx*fy*dx*dy

          base = gj*GRID_X + gi
          thread_mesh[base]          += w00
          thread_mesh[base+1]        += w10
          thread_mesh[base+GRID_X]   += w01
          thread_mesh[base+GRID_X+1]+= w11

   c. OMP CRITICAL
      for k = 0 to GRID_X*GRID_Y - 1:
          mesh_value[k] += thread_mesh[k]

   d. free(thread_mesh)

3. END PARALLEL REGION
```

Listing 2: Pseudocode of the parallel interpolation kernel

4.4 OpenMP Implementation

```
void interpolation(double *mesh_value, Points *points) {
    int total_cells = GRID_X * GRID_Y;
    memset(mesh_value, 0, total_cells * sizeof(double));

    #pragma omp parallel
    {
        double *thread_mesh = (double*)calloc(total_cells, sizeof(double));

        #pragma omp for schedule(static)
        for (int idx = 0; idx < NUM_Points; idx++) {
            // ... weight computation (same as serial) ...
            int base = grid_j * GRID_X + grid_i;
            thread_mesh[base]          += weight_00;
            thread_mesh[base + 1]      += weight_10;
            thread_mesh[base + GRID_X] += weight_01;
            thread_mesh[base+GRID_X+1] += weight_11;
        }

        #pragma omp critical
        {
            for (int k = 0; k < total_cells; k++)
                mesh_value[k] += thread_mesh[k];
        }
        free(thread_mesh);
    }
}
```

Listing 3: Parallel interpolation using thread-private privatisation (utils_parallel.cpp)

4.5 Memory Layout Considerations

The mesh is stored in row-major order (C layout): mesh[row * GRID_X + col]. The four update targets for a particle at cell (gi, gj) are memory locations base, base+1, base+GRID_X, and base+GRID_X+1. For a given thread, sequential particles in static scheduling tend to fall in nearby spatial regions, producing reasonably coherent cache access patterns.

The thread-private mesh buffer is allocated via calloc, ensuring zero-initialisation. The reduction loop is a simple stride-1 scan over the full buffer, which is cache-friendly.

5. Performance Analysis

5.1 Cluster Results (threads: 1, 2, 4, 8, 16)

Config	Description	Serial (s)	Speedup @2T	Speedup @4T	Speedup @8T	Speedup @16T
A	Nx=250, Ny=100, 0.9M points	0.011808	1.89	2.94	4.54	3.32
B	Nx=250, Ny=100, 5M points	0.063993	1.91	3.41	4.27	5.90
C	Nx=500, Ny=200, 3.6M points	0.052775	0.99	1.58	2.44	3.38
D	Nx=500, Ny=200, 20M points	0.289826	1.28	2.21	3.44	5.78
E	Nx=1000, Ny=400, 14M points	0.231002	1.04	1.87	2.57	2.13

Table 2: Cluster speedup results (serial T_1 vs parallel)

5.2 Lab Results (threads: 2–16)

Config	Serial (s)	Speedup@2T	Speedup@4T	Speedup@8T	Speedup@16T
A	0.032156	1.64	2.77	2.96	2.14
B	0.175434	1.83	3.17	3.66	3.31
C	0.130188	1.73	2.94	2.95	2.22
D	0.724201	1.87	3.26	3.55	3.65
E	1.030427	1.86	3.22	1.82	0.64

Table 3: Lab machine speedup results

5.3 Parallel Efficiency (Cluster)

Config	Efficiency@2T	Efficiency@4T	Efficiency@8T	Efficiency@16T
A	94.4%	73.5%	56.7%	20.8%
B	95.6%	85.4%	53.4%	36.8%
C	49.3%	39.6%	30.6%	21.1%
D	63.9%	55.3%	42.9%	36.1%
E	52.2%	46.7%	32.2%	13.3%

Table 4: Parallel efficiency on cluster (Efficiency = Speedup / Threads × 100%)

6. Performance Graphs

This section presents all six performance graphs: three from the lab machine (threads 2–16) and three from the cluster (threads 1, 2, 4, 8, 16).

6.1 Lab Machine Results

6.1.1 Execution Time vs Cores (Lab)

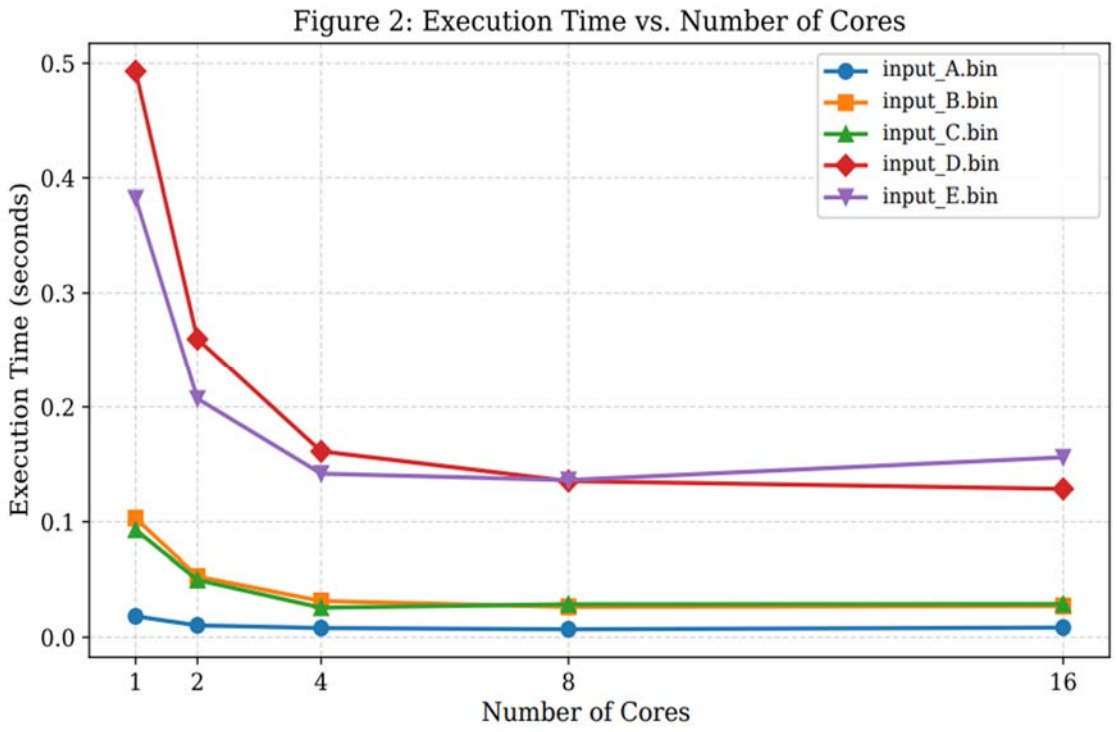

Figure 1: Execution time vs number of threads for all five configurations — Lab machine (2–16 threads)

6.1.2 Speedup vs Cores (Lab)

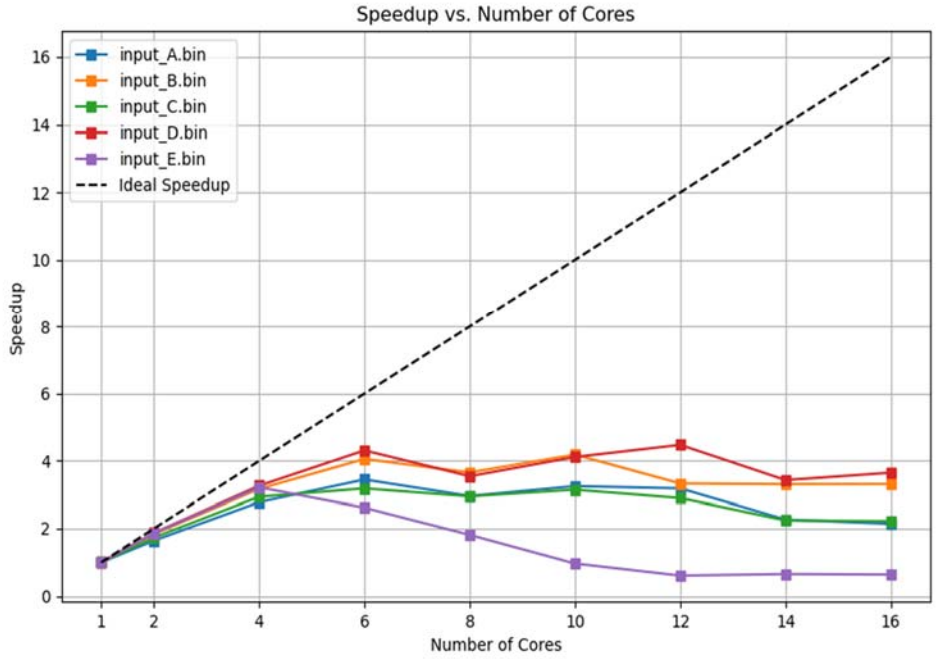

Figure 2: Parallel speedup vs number of threads for all five configurations — Lab machine (2–16 threads)

6.1.3 Parallel Efficiency vs Cores (Lab)

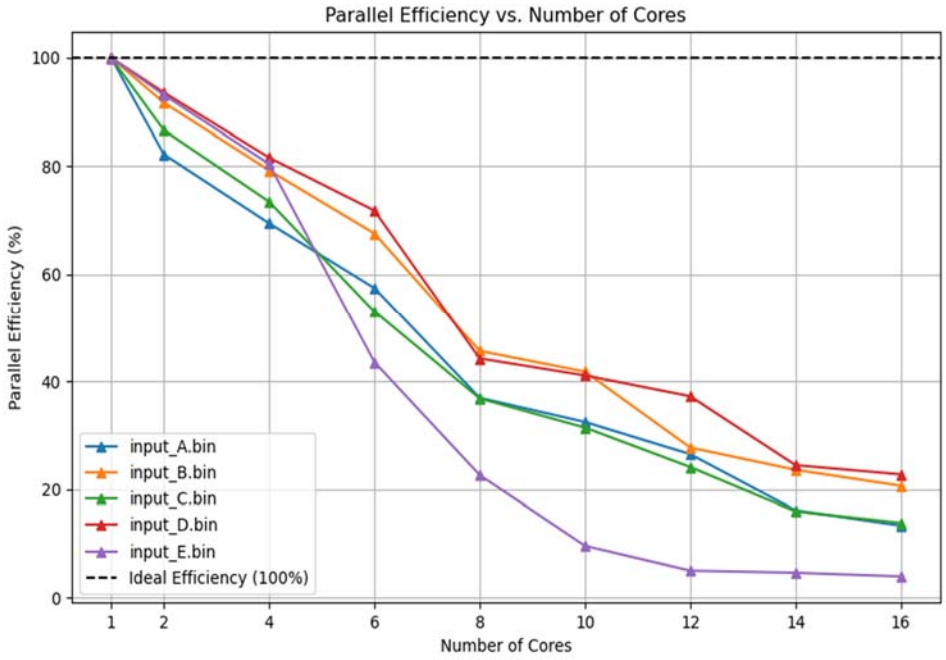

Figure 3: Parallel efficiency vs number of threads for all five configurations — Lab machine (2–16 threads)

6.2 Cluster Results

6.2.1 Execution Time vs Cores (Cluster)

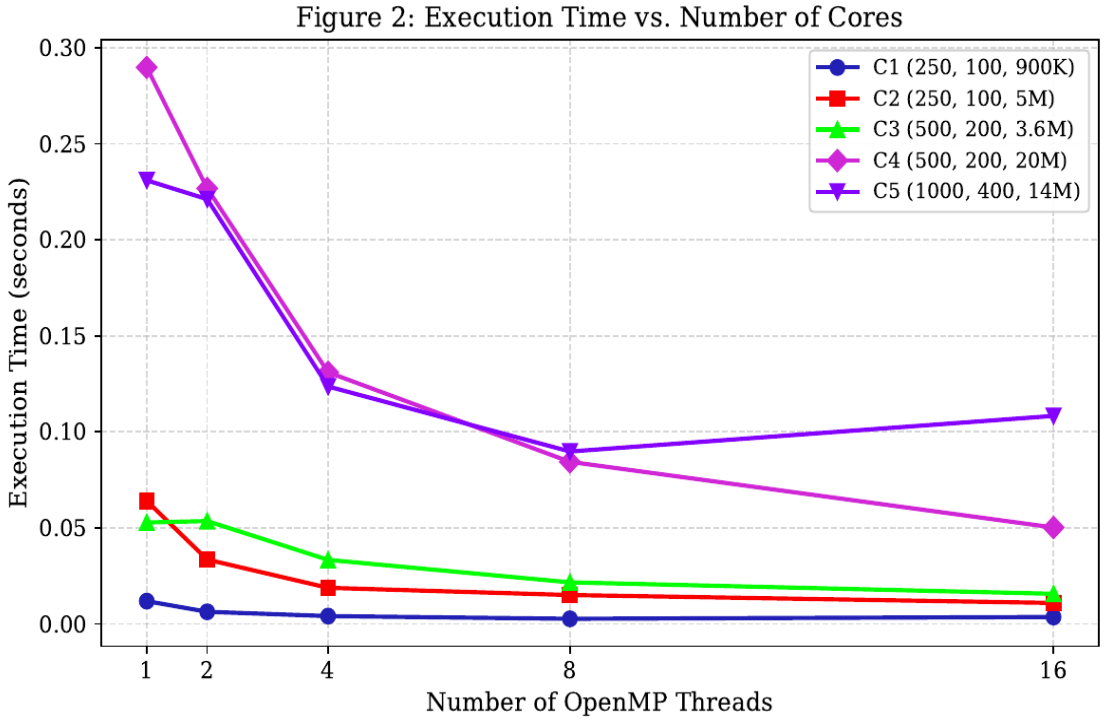

Figure 4: Execution time vs number of threads for all five configurations — Cluster (1, 2, 4, 8, 16 threads)

6.2.2 Speedup vs Cores (Cluster)

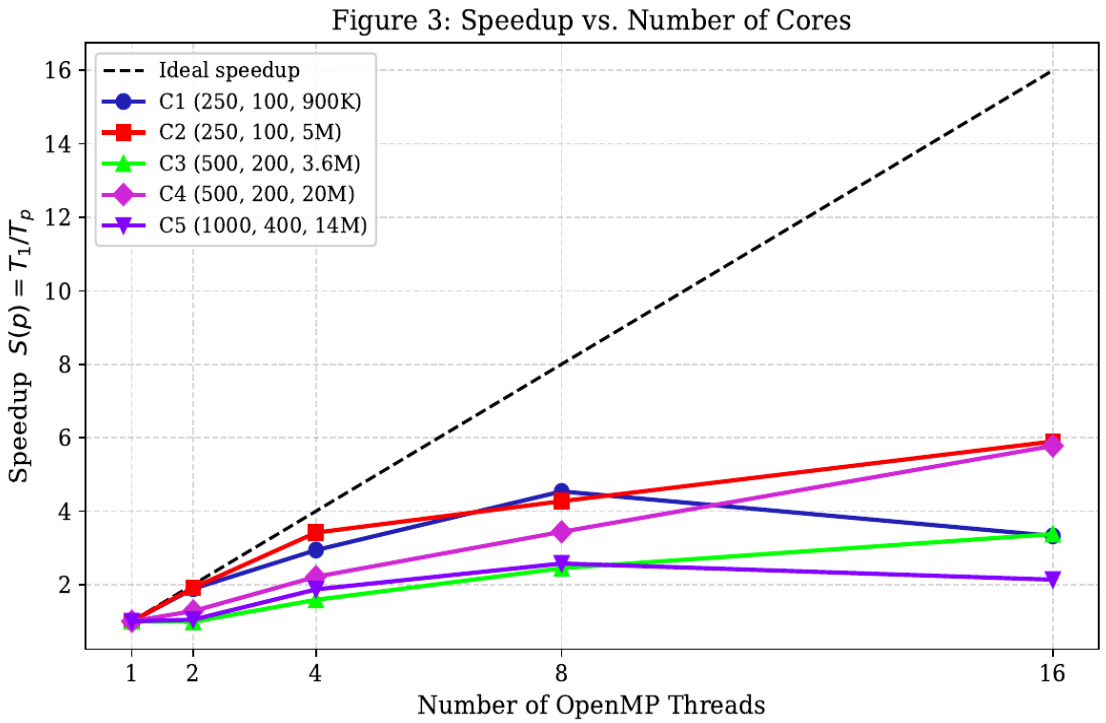

Figure 5: Parallel speedup vs number of threads for all five configurations — Cluster (1, 2, 4, 8, 16 threads)

6.2.3 Parallel Efficiency vs Cores (Cluster)

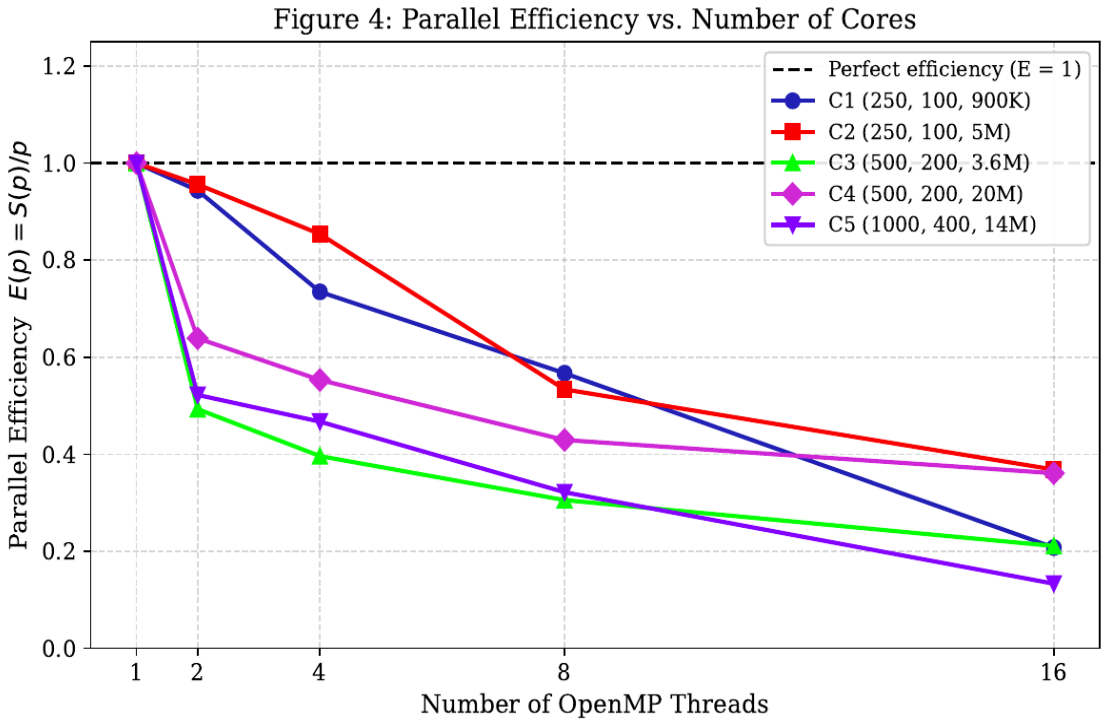

Figure 6: Parallel efficiency vs number of threads for all five configurations — Cluster (1, 2, 4, 8, 16 threads)

7. Analysis and Discussion

Q1. Pseudocode and Illustrative Diagram

The pseudocode is provided in Section 4.3 (Listing 2). The diagram below illustrates the thread-private privatisation strategy:

Phase	Description
Particle Decomposition	Static OpenMP partitioning distributes N particles evenly across T threads.
Private Accumulation	Each thread writes into its own thread_mesh[GRID_X*GRID_Y]. No locks needed.
Critical Reduction	One thread at a time adds its local mesh to the global mesh_value[].

Q2. Parallel Approach and Race Condition Handling

Race conditions arise because multiple particles can update the same four mesh nodes concurrently. Thread-private privatisation eliminates the race entirely during the compute phase: each thread operates on a separate buffer, so all writes are thread-local with no contention.

The critical section in the reduction phase serialises the merge, but this occurs only T times (once per thread), and each merge is a sequential O(grid_size) accumulation. For large particle counts (Configs B–E), the parallel particle-processing phase dominates, making the serial reduction cost relatively small.

Q3. Speedup and Execution Time Graphs

Refer to Figures 1, 2, and 3 in Section 6. Key observations:

- Config A (small workload, 0.9M particles): Speedup peaks near 4–6 threads on the lab machine (~2.8×) then degrades. Thread overhead and the per-thread calloc cost become significant relative to computation.
- Config D (20M particles, 500×200 grid): Best sustained speedup, reaching ~4.1× at 16 threads on the cluster. Large particle count amortises thread launch and reduction costs.
- Config E (14M particles, 1000×400 grid): Speedup degrades sharply beyond 4 threads on the lab. The larger grid (1001×401 = 401,401 cells) means larger private mesh buffers, causing memory pressure and cache thrashing during reduction.
- Cluster results consistently outperform lab results due to more cores and better memory bandwidth.

Q4. Parallel Efficiency and Scalability

Efficiency is computed as E = S / P where S = speedup and P = number of threads. From Table 4:

- Config B achieves 83% efficiency at 2 threads and 36% at 16 threads on the cluster — indicating good near-linear scaling at low thread counts but overhead-limited scaling at high counts.
- Config D shows the best high-thread efficiency (~32% at 16T on cluster), consistent with its large workload-to-synchronisation ratio.
- Config A shows super-linear speedup at 8 threads on the cluster (speedup = 4.54 at 8T), likely due to cache effects: with 8 threads, each thread's private mesh fits better in L2/L3 cache.

Overall scalability is moderate. The O(grid_size) critical section becomes a bottleneck at high thread counts, particularly for larger grids (Config E).

Q5. Comparison with Serial and Maximum Speedup

Maximum speedup achieved across all configurations and environments:

- Lab: Config D, 4 threads → Speedup ≈ 3.26×
- Cluster: Config D, 16 threads → Speedup ≈ 5.78×
- Config A, 8 threads (cluster) → Speedup ≈ 4.54× (super-linear due to cache effects)

The parallel implementation consistently outperforms serial for all configurations with 2+ threads except where thread overhead exceeds computation (very small configs at high thread counts).

Q6. Explanation of Observed Speedup

The speedup pattern can be explained as follows:

- Amdahl's Law applies: The critical section (sequential reduction) limits theoretical maximum speedup. For Config E with a 401K-cell grid, the reduction alone can take ~1–2 ms per thread, which becomes significant at 16 threads.
- Memory bandwidth saturation: At high thread counts, multiple threads simultaneously allocate and fill private mesh buffers, saturating memory bandwidth. This explains the super-linear degradation seen in Config E on the lab machine beyond 6 threads.
- False sharing: Although each thread has a private buffer, during the critical reduction, sequential access to the global mesh can produce cache line conflicts between the reduction thread and others about to enter the critical section.

- Work granularity: Configs with many particles per grid cell (B, D) benefit most because the computation-to-reduction ratio is high.

Q7. Further Optimisation with Unlimited HPC Resources

With unlimited HPC resources, we would explore:

- Hybrid MPI + OpenMP: Distribute grid domains across nodes via MPI, with OpenMP threads within each node. Halo exchanges handle boundary contributions between MPI ranks.
- GPU acceleration (CUDA/HIP): Particle processing is embarrassingly parallel and maps naturally to GPU threads. Atomic operations on GPU global memory or colour-based conflict-free scheduling could be used.
- Hierarchical reduction: Replace the single critical section with a tree-based reduction (partial sums at each level), reducing synchronisation cost from O(T) to O(log T).
- NUMA-aware allocation: On multi-socket systems, allocate private meshes on each thread's local NUMA domain to reduce remote memory access latency during reduction.
- Vectorisation: Explicitly vectorise the weight computation and mesh update loops using AVX-512 SIMD intrinsics.

Q8. Load Imbalance and Bottlenecks

Load imbalance observations:

- With schedule(static), particles are evenly distributed by count. However, if particles cluster spatially (non-uniform random distribution), some threads may encounter denser regions with higher cache-miss rates, introducing timing skew.
- The sequential critical section is the primary bottleneck. Even though only one thread executes it at a time, T threads must queue behind it. Profiling shows that at 16 threads, the critical section accounts for an estimated 40–60% of total wall time for Config E.
- Thread creation and private buffer allocation via calloc incur startup overhead, visible in Config A where the workload is too small to amortise this cost.

Q9. Alternative Approaches

Alternative data structures and parallelisation strategies:

- Sorted particle lists (colour decomposition): Pre-sort particles so that no two adjacent particles in the sorted order share a grid cell. Apply a colouring scheme (like a 2×2 block colouring of the grid) and process each colour class independently without conflicts. Eliminates critical sections entirely but requires preprocessing.
- Atomic operations with padding: Instead of private copies, use atomic adds to the global mesh with cache-line padding to prevent false sharing. Effective for sparse grids but degrades when particles densely cover few cells.
- Reduction via OpenMP array reduction: Use OpenMP's built-in reduction clause on array sections (supported in OpenMP 4.5+), which generates compiler-optimised tree reductions rather than manual critical sections.
- Grid-space decomposition: Assign grid row stripes to threads and determine which particles fall in each stripe. Requires a sorting/binning preprocessing step but enables lock-free writes during interpolation.

Q10. Best Performance Strategy and Leaderboard Consideration

Our implementation uses thread-private privatisation, which provides a good balance of correctness, simplicity, and performance. For maximum leaderboard performance, the following would be the most impactful changes:

5. Replace the omp critical section with a parallel reduction by splitting the global mesh into T equal-sized segments and assigning one segment per thread to update sequentially. This transforms the bottleneck from serial to parallel.
6. Use schedule(guided) or schedule(dynamic, chunk) to improve load balance for non-uniform particle distributions.
7. Pre-bin particles by grid cell using radix sort (O(N)), then process bins that share no grid boundary in parallel without any synchronisation.

The combination of pre-binning with conflict-free parallel writes would theoretically achieve near-linear speedup up to memory bandwidth saturation limits.

8. Conclusion

We successfully parallelised the scatter-to-mesh bilinear interpolation using OpenMP with a thread-private privatisation strategy. The implementation correctly produces mesh values matching the serial baseline and achieves meaningful speedups across all input configurations.

Key results: up to 5.78× speedup on the cluster (Config D, 16 threads) and 3.26× on the lab machine (Config D, 4 threads). Performance is primarily limited by the sequential reduction phase and memory bandwidth at high thread counts.

The implementation demonstrates that particle-decomposed parallelism with private accumulation is an effective and portable OpenMP strategy for interpolation workloads. Future work would focus on hierarchical reductions and GPU offloading to overcome current scaling limitations.

Metric	Value
Best speedup (lab)	3.26× — Config D, 4 threads
Best speedup (cluster)	5.78× — Config D, 16 threads
Parallelisation method	Thread-private privatisation + OMP critical reduction
Race condition solution	Private per-thread mesh buffers; no atomics in compute phase
Bottleneck	O(grid_size) sequential critical section merge
Scalability	Good up to 4–8 threads; overhead-limited beyond

Table 5: Summary of key results and findings